Theoretical Framework for Trajectory-Based Metaheuristics

Wind Farm Layout Optimization and the Dynamic Drone Routing Problem

MetaLab Suite

Department of Operations Research

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Abstract

This document establishes the theoretical foundations of the MetaLab Suite, an educational repository devoted to *trajectory-based metaheuristics* for combinatorial and continuous optimisation. We develop, in full mathematical detail, two flagship problems. **Part I** addresses the *Wind Farm Layout Optimization* (WFLO) problem: given a terrain and a wind resource, where should turbines be placed so as to maximise the expected power output while mitigating the aerodynamic *wake effect*? We derive the objective function P_{total}, state the topographical and engineering constraints, and present a simplified *Jensen* wake model, before arguing why the resulting landscape is strongly multimodal and therefore calls for trajectory-based search. **Part II** addresses the *Dynamic Drone Routing Problem* (DDRP) in the spirit of Dorling et al. (2016): a fleet of unmanned aerial vehicles serves delivery orders that are revealed in real time, subject to a strict payload capacity Q and a battery budget whose consumption scales with both payload and travelled distance. We formulate the objective (minimise distance plus unserved-order penalties) and the associated constraints, extend the model to the dynamic setting, and close with the design of trajectory metaheuristics for both problems.

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1 Introduction

1.1 Scope and motivation

Many real-world planning problems are *black-box*, *constrained* and *multimodal*: the objective cannot be differentiated analytically, feasible regions are riddled with holes, and the landscape contains a combinatorial profusion of local optima. Two representative examples motivate this document: the siting of wind turbines on a complex terrain (Part I) and the routing of delivery drones under real-time demand (Part II).

Trajectory-based metaheuristics—also called *single-solution* or *local-search* metaheuristics—are particularly well suited to such problems. Rather than maintaining a population, they evolve a single incumbent solution $x^0, x^1, \dots, x^T$ by iteratively exploring a *neighbourhood* $\mathcal{N}(x)$ and applying an acceptance rule that may temporarily worsen the objective so as to escape local optima.

1.2 Trajectory-based metaheuristics: common skeleton

Formally, given a finite or continuous *search space* $\mathcal{X}$, an objective $f : \mathcal{X} \to \mathbb{R}$ (to be minimised or maximised), and a neighbourhood operator $\mathcal{N} : \mathcal{X} \to 2^{\mathcal{X}}$, a generic trajectory method proceeds as in Algorithm 1.

Algorithm 1 Generic trajectory-based metaheuristic

Initialise $x \leftarrow x^0 \in \mathcal{X}$ (e.g. at random or greedily)
$x^\star \leftarrow x$ ▷ best solution found so far
while stopping criterion not met **do**
    Generate a candidate $x' \in \mathcal{N}(x)$
    if Accept(x, x', state) **then**
        $x \leftarrow x'$
    end if
    if $f(x')$ improves $f(x^\star)$ **then**
        $x^\star \leftarrow x'$
    end if
    Update the search *state* (temperature, tabu tenure, . . .)
end while
return $x^\star$

Three instantiations are used throughout the suite and differ only in the acceptance rule:

- **Simulated annealing (SA)** accepts a worsening move with probability $\exp(-\Delta/T)$, where $\Delta > 0$ is the objective degradation and T a slowly decreasing temperature [4].
- **Tabu search (TS)** always moves to the best admissible neighbour but forbids recently visited configurations through a short-term memory (the *tabu list*), with an aspiration criterion to override the tabu status on sufficiently good moves [5].
- **Iterated local search (ILS)** alternates a local-search descent with a stochastic *perturbation* of the incumbent, accepting the result by a simple criterion [6].

A crucial property for Part II is that trajectory methods are *anytime*: they may be interrupted at any moment and still return the best solution found so far, which is exactly what a real-time dispatcher needs.

1.3 Notation

Table 1: Global notation used in both parts.

Symbol	Meaning
$\mathcal{X}$	search space (problem-dependent encoding)
$f(\cdot)$	objective function of the metaheuristic
$\mathcal{N}(x)$	neighbourhood of solution x
$x^\star$	best solution found during a run
T	temperature (SA)
$\mathcal{T}$	tabu list (TS)
N	number of turbines (Part I)
Q	payload capacity of a drone (Part II)
B	battery energy budget of a drone (Part II)

2 Part I — Wind Farm Layout Optimization (WFLO)

2.1 Problem description

A wind farm occupies a domain $\Omega \subset \mathbb{R}^2$ (the available land, possibly irregular and with an elevation profile). We must choose the positions

$$X = (x_1, x_2, \ldots, x_N), \qquad x_j = (x_j^{(1)}, x_j^{(2)}) \in \Omega,$$

of N identical turbines so that the farm's electrical production is as large as possible over a reference year. The fundamental difficulty is the *wake effect*: an upwind turbine extracts kinetic energy from the flow and leaves behind it a region of reduced wind speed and increased turbulence. A turbine standing in that wake therefore produces less than it would in free-stream conditions. Pushing turbines apart reduces wake losses but consumes land and increases collector-system and civil costs; clustering them together saves land but intensifies interference. The optimal design must balance these two effects, which is what makes the problem non-trivial.

2.2 The wake effect

Let the wind travel in direction θ (the bearing it blows towards), with unit vector $\mathrm{n} = (\cos\theta, \sin\theta)$ and its normal $\mathrm{n}^\perp = (-\sin\theta, \cos\theta)$. For two turbines i and j, define the *along-wind* and *cross-wind* separations

$$x_{ij} = (x_j - x_i) \cdot \mathrm{n}, \qquad y_{ij} = \left|(x_j - x_i) \cdot \mathrm{n}^\perp\right|. \tag{1}$$

A deficit is created behind turbine i only if $x_{ij} > 0$ (turbine j is downwind of i) and if j lies inside the expanding wake.

2.3 Simplified Jensen wake model

The Jensen (or Katić–Jensen) model [2, 3] assumes a *top-hat* velocity profile: inside the wake the wind speed is uniform, and the wake expands linearly with downstream distance at a rate governed by the decay constant k. Its radius at downstream distance x from a rotor of diameter D is

$$R_w(x) = \frac{D}{2} + k\,x. \tag{2}$$

When turbine j is partially or fully immersed in the wake of i, the fractional velocity deficit is

$$\delta_{ij} = \left(1 - \sqrt{1 - C_T}\right) \left(\frac{D}{D + 2k\, x_{ij}} \right)^2, \qquad \delta_{ij} = 0 \text{ if } x_{ij} \le 0 \text{ or } y_{ij} > R_w(x_{ij}), \tag{3}$$

where $C_T \in (0, 1)$ is the thrust coefficient of the turbine. The factor $1 - \sqrt{1 - C_T}$ equals twice the axial induction factor a, so Eq. (3) has a clear physical interpretation: the maximum deficit immediately behind the rotor is $2a$, and it decays as the wake spreads.

For a given wind scenario the total deficit at turbine j is obtained by superposing the contributions of all upwind turbines, most commonly in *root-sum-square* form,

$$U_j = U_\infty \left(1 - \sqrt{\sum_{i \in \mathcal{I}_j} \delta_{ij}^2} \right)_+, \qquad \mathcal{I}_j = \{\, i : x_{ij} > 0 \ \wedge \ y_{ij} \le R_w(x_{ij}) \,\}, \tag{4}$$

with U_∞ the free-stream speed and $(\cdot)_+ = \max(\cdot, 0)$ to keep the effective speed non-negative. Linear superposition $U_j = U_\infty(1 - \sum_i \delta_{ij})$ is a common alternative; both are supported by the reference implementation. The Jensen model is deliberately simple. It neglects turbulence anisotropy, ground effects and yaw, yet it reproduces the qualitative behaviour that dominates the optimisation and keeps the objective cheap enough for the tens of thousands of evaluations a metaheuristic requires.

2.4 Turbine power curve

The electrical power produced by a single turbine as a function of wind speed u is modelled by the standard piecewise *power curve*. With cut-in speed u_{in}, rated speed u_r, cut-out speed u_{out} and rated power P_r,

$$P(u) = \begin{cases} 0, & u < u_{\text{in}} \text{ or } u \ge u_{\text{out}}, \\ P_r \left(\dfrac{u - u_{\text{in}}}{u_r - u_{\text{in}}} \right)^3, & u_{\text{in}} \le u < u_r, \\ P_r, & u_r \le u < u_{\text{out}}. \end{cases} \tag{5}$$

The cubic regime reflects the fact that the kinetic power of the wind scales with the cube of its speed, $P_{\text{wind}} = \tfrac{1}{2}\rho A u^3$, where ρ is air density and A the rotor swept area.

2.5 Wind resource: the discretised wind rose

The wind climate is summarised by a *wind rose*: a finite set of S scenarios $\{(\theta_s, U_s^\infty, p_s)\}_{s=1}^S$, where θ_s is a direction, U_s^∞ a representative speed and $p_s \in [0, 1]$ its relative frequency, with $\sum_{s=1}^S p_s = 1$. The rose is typically obtained by binning historical anemometry data in direction and speed.

2.6 Objective function: expected power P_{total}

For scenario s, applying (4) and (5) to every turbine gives the farm's instantaneous power

$$P_{\text{total}}^{(s)}(X) = \sum_{j=1}^{N} P\big(U_j^{(s)}(X)\big), \tag{6}$$

and the *expected* power output of the layout is the probability-weighted average over the rose,

$$\boxed{\bar{P}(X) = \sum_{s=1}^{S} p_s\, P_{\text{total}}^{(s)}(X) = \sum_{s=1}^{S} p_s \sum_{j=1}^{N} P\Big(U_{\infty,s} \Big(1 - \sqrt{\textstyle\sum_{i \in \mathcal{I}_j} \delta_{ij}^2} \Big)_+ \Big)}. \tag{7}$$

Multiplying by the number of hours in a year, $T_{\text{year}} = 8760$ h, yields the *Annual Energy Production* $\text{AEP}(X) = T_{\text{year}}\,\bar{P}(X)$. Because turbines may be assigned a hub height $H_j = h(x_j) + H_0$ that follows the terrain elevation h, the effective speeds in (7) can be evaluated at the local hub height, although we keep the notation compact here. The WFLO problem is then

$$\text{(WFLO)} \qquad \max_{X \in \mathcal{F}} \ \bar{P}(X), \tag{8}$$

where $\mathcal{F}$ is the feasible set defined next.

2.7 Topographical and engineering constraints

The feasible set $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2 \cap \mathcal{F}_3 \cap \mathcal{F}_4 \cap \mathcal{F}_5$ encodes the physical and regulatory limits.

($\mathcal{F}_1$) Land availability / boundary clearance. Every turbine, including its rotor, must lie inside the buildable domain Ω:

$$\text{dist}(x_j, \partial\Omega) \ \geq \ R = D/2, \qquad j = 1, \ldots, N. \tag{9}$$

On real terrain Ω excludes water bodies, settlements, roads and other protected areas.

($\mathcal{F}_2$) Exclusion zones. A collection of forbidden polygons $\{Z_m\}_{m=1}^{M}$ (buildings, radar cones, archaeological sites) must be avoided:

$$x_j \notin \bigcup_{m=1}^{M} Z_m, \qquad j = 1, \ldots, N. \tag{10}$$

($\mathcal{F}_3$) Minimum inter-turbine spacing. To limit wake interference and mechanical loading, turbines must keep a centre-to-centre distance of at least a multiple γ of the rotor diameter:

$$\|x_i - x_j\| \ \geq \ s_{\min} = \gamma D, \qquad 1 \leq i < j \leq N, \tag{11}$$

with typical $\gamma \in [3, 8]$. Eq. (11) makes $\mathcal{F}$ *non-convex*: it removes an open neighbourhood around each turbine, effectively carving holes in the domain.

($\mathcal{F}_4$) Terrain and slope. Turbine foundations require limited ground slope and, for road access, acceptable gradients between positions. A common surrogate bounds the elevation difference relative to the horizontal separation,

$$\frac{|h(x_i) - h(x_j)|}{\|x_i - x_j\|} \ \leq \ \sigma_{\max}, \qquad 1 \leq i < j \leq N, \tag{12}$$

and hub heights are constrained to a permissible band $H_{\min} \leq h(x_j) + H_0 \leq H_{\max}$.

($\mathcal{F}_5$) Discrete/engineering constraints. Optional restrictions such as a maximum number of turbines along a line, alignment rules, or a grid-snapping of positions to an accessible lattice.

2.8 Why the landscape is multimodal, and why trajectory search fits

The WFLO objective (7) is, for several independent reasons, a pathological landscape for gradient-based methods.

Proposition 1 (Structural difficulties)**.** *The WFLO objective $\bar{P}$ over $\mathcal{F}$ exhibits the following properties.*

1. ***Non-convexity and disconnectedness.*** *The spacing constraints* (11) *render the feasible set non-convex; it is a union of many disjoint regions in $\mathbb{R}^{2N}$.*

2. ***Non-separability.*** *Because of the wake terms δ_{ij}, the contribution of turbine j depends on the positions of all turbines upwind of it. The objective does not decompose into a sum of per-turbine terms, so turbines cannot be optimised independently.*

3. ***Nonsmoothness.*** *The indicator condition $y_{ij} \le R_w(x_{ij})$ in (3) introduces a discontinuity in the first derivative of $\bar{P}$ whenever a turbine enters or leaves a wake cone; likewise the $(\cdot)_+$ and the piecewise power curve are non-differentiable at their kinks.*

4. ***Combinatorial multimodality.*** *Each of the $\binom{N}{2}$ wake interactions contributes a local basin, so the number of local optima grows rapidly with N. Good layouts typically arrange turbines in staggered patterns aligned with the dominant wind direction, and distinct wind-rose structures give rise to competing, mutually exclusive configurations.*

Classical nonlinear programming solvers stall on the first local optimum and cannot cross the spacing barriers of (F3). This is precisely the regime in which *trajectory-based* metaheuristics excel: they treat $\bar{P}$ as a black box, they navigate non-convex feasible sets through clever encodings and penalties, and their acceptance rules (Metropolis, tabu tenure, perturbation) let them escape the numerous local optima catalogued in Proposition 1.

Encoding and neighbourhood. Because the natural variables are continuous, trajectory methods operate directly on the coordinate vector $X \in \mathbb{R}^{2N}$. A standard neighbourhood perturbs a single turbine,

$$\mathcal{N}_{\text{move}}(X) = \{\, X + e_j \otimes \xi \;:\; 1 \le j \le N,\ \xi \sim \mathcal{N}(0, \sigma^2 I_2) \,\}, \tag{13}$$

optionally combined with *swap* moves (exchange two turbines) and *block* moves (translate a subset of turbines). Infeasible candidates are handled either by rejection, by a repair operator, or by a penalty $\lambda \sum_{i<j} [s_{\min} - \|x_i - x_j\|]_+$.

Reference algorithm. Algorithm 2 instantiates the skeleton with a Metropolis acceptance and a geometric cooling schedule $T_{t+1} = \alpha T_t$; this is the reference metaheuristic against which the baseline hill climber of the repository is compared.

Algorithm 2 Simulated annealing for WFLO (maximisation of $\bar{P}$)

$X \leftarrow$ RandomFeasibleLayout(N, Ω)
$X^\star \leftarrow X$; $T \leftarrow T_0$
while $T > T_{\text{end}}$ **and** budget remains **do**
  repeat
    Draw $X' \in \mathcal{N}_{\text{move}}(X)$ and repair/clip to $\mathcal{F}$
    $\Delta \leftarrow \bar{P}(X') - \bar{P}(X)$
    if $\Delta > 0$ **or** rand() $< \exp(\Delta/T)$ **then**
      $X \leftarrow X'$
      if $\bar{P}(X) > \bar{P}(X^\star)$ **then**
        $X^\star \leftarrow X$
      end if
    end if
  until L iterations completed
  $T \leftarrow \alpha T$
end while
return $X^\star$

3 Part II — Dynamic Drone Routing Problem (DDRP)

3.1 Motivation and related work

Unmanned aerial vehicles (drones) are increasingly used for last-mile parcel delivery. Dorling et al. [1] introduced and studied the *Vehicle Routing Problem for drone delivery*, in which a fleet of quad-rotor drones, launched from a depot, serves a set of customers under hard limits on payload weight and battery energy. Their model demonstrates that the cost of a delivery is governed jointly by *distance* and by *payload-dependent energy* consumption, so the routing and the loading decisions are inseparable.

In the *Dynamic* Drone Routing Problem (DDRP) studied here, delivery orders are not known in advance: they are revealed in *real time* over a planning horizon. The dispatcher must continuously reassign drones, implying a trade-off between exploiting the current plan and re-optimising as new information arrives. Dynamic routing of this kind generalises the static vehicle-routing problem and is NP-hard [9]; its online nature makes it even more challenging [10].

3.2 Sets, parameters and decision variables

Table 2: DDRP data and decision variables.

Symbol	Meaning
$\mathcal{K} = \{1, \dots, K\}$	set of drones
$\mathcal{I} = \{1, \dots, n\}$	set of delivery orders (customers)
0	the depot (*departure/return node*)
$V = \{0\} \cup \mathcal{I}$	node set
d_{ij}	travel distance between nodes i, j
w_i	payload (parcel weight) of order i
Q	payload capacity of a drone
B	usable battery energy of a drone
a	empty-drone energy consumption per unit distance
b	payload energy sensitivity per unit distance
v	cruise speed (distance per unit time)
τ	service (hover/landing) time per order
λ	penalty for an unserved order
$x_{ijk} \in \{0, 1\}$	1 if drone k travels directly from i to j
$y_{ik} \in \{0, 1\}$	1 if drone k serves order i
$q_{ijk} \geq 0$	payload carried by drone k on arc (i, j)
$t_{ik} \geq 0$	(auxiliary) position of node i in route k
$z_k \in \{0, 1\}$	1 if drone k is used

3.3 Payload-dependent battery model

A multi-rotor's power requirement increases with its total mass, hence with the payload it carries. Following the linearisation used by Dorling et al. [1], the *energy consumed per unit distance* by a drone carrying a payload q is

$$e(q) = a + b\,q, \qquad a \geq 0,\ b \geq 0, \tag{14}$$

so that the energy spent traversing arc (i, j) with payload q is $e(q)\,d_{ij}$. The two constants capture the empty-drone consumption a and the extra consumption caused by carrying mass $(b\,q)$. This dependency is what makes DDRP different from a pure distance-minimisation problem: a heavy parcel carried far consumes disproportionately more battery.

3.4 Objective function

Let $y_{ik} = \sum_{j \in V} x_{ijk}$ indicate whether drone k serves order i. The problem trades off three quantities: total travelled distance, total energy consumed, and the number of orders left unserved. The primary objective requested here—*minimise distance and unserved penalties*—reads

$$\boxed{\min \; Z(X) = \underbrace{\sum_{k \in \mathcal{K}} \sum_{i \in V} \sum_{j \in V} d_{ij}\, x_{ijk}}_{\text{total travelled distance}} + \lambda \underbrace{\sum_{i \in \mathcal{I}} \Big(1 - \sum_{k \in \mathcal{K}} y_{ik}\Big)}_{\text{unserved orders}}} \tag{15}$$

where $\lambda \gg \max_{ij} d_{ij}$ is chosen large enough that serving an order is always preferable to abandoning it, yet finite so that genuinely unreachable orders under the battery budget are penalised rather than rendering the instance infeasible. If energy is to be priced explicitly (as in [1]), the objective is augmented with $c_e \sum_k \sum_{i,j} (a + b\, q_{ijk})\, d_{ij}\, x_{ijk}$, where c_e is the cost per unit of energy; we keep the two-term form (15) as the canonical teaching objective.

3.5 Constraints

The following constraints are *strict*: no plan may violate them.

(C1) Unique assignment and unserved accounting. Each order is served by at most one drone,

$$\sum_{k \in \mathcal{K}} y_{ik} \;\le\; 1, \qquad i \in \mathcal{I}. \tag{16}$$

(C2) Flow conservation and route structure. For every served node the in-degree equals the out-degree, and each used drone leaves and re-enters the depot,

$$\sum_{j \in V} x_{ijk} = \sum_{j \in V} x_{jik} = y_{ik}, \quad i \in \mathcal{I},\; k \in \mathcal{K}; \qquad \sum_{j \in \mathcal{I}} x_{0jk} = \sum_{i \in \mathcal{I}} x_{i0k} = z_k, \quad k \in \mathcal{K}. \tag{17}$$

(C3) Payload capacity. The payload flowing into a customer equals the payload flowing out plus the parcel delivered there; the depot loads each drone with exactly the parcels it will deliver:

$$\sum_{j \in V} q_{ijk} - \sum_{j \in V} q_{jik} = -\, w_i\, y_{ik}, \qquad i \in \mathcal{I},\; k \in \mathcal{K}, \tag{18}$$

$$\sum_{j \in V} q_{0jk} = \sum_{i \in \mathcal{I}} w_i\, y_{ik}, \qquad k \in \mathcal{K}, \tag{19}$$

and the capacity limit holds on every arc and for the whole load,

$$0 \;\le\; q_{ijk} \;\le\; Q\, x_{ijk}, \quad i, j \in V,\; k \in \mathcal{K}; \qquad \sum_{i \in \mathcal{I}} w_i\, y_{ik} \;\le\; Q\, z_k, \quad k \in \mathcal{K}. \tag{20}$$

(C4) Battery budget. The energy consumed along the whole route, with consumption scaling with payload (14) and distance, must not exceed the usable battery energy,

$$\sum_{i \in V} \sum_{j \in V} (a + b\, q_{ijk})\, d_{ij}\, x_{ijk} \;\le\; B\, z_k, \qquad k \in \mathcal{K}. \tag{21}$$

In practice one reserves a safety margin B_{res} for wind, hovering and landing, replacing B by $B - B_{\text{res}}$.

(C5) Subtour elimination. Auxiliary order variables with a big-M bound forbid disconnected subtours,

$$t_{ik} - t_{jk} + n\, x_{ijk} \;\le\; n - 1, \qquad i, j \in \mathcal{I},\; i \neq j,\; k \in \mathcal{K}. \tag{22}$$

(C6) Release times (dynamic information). An order i with release time r_i may be served only once it is known. Let t^a_{ik} denote the arrival time at order i. Then

$$y_{ik} = 1 \;\Rightarrow\; t^a_{ik} \;\geq\; r_i, \tag{23}$$

which we linearise via $t^a_{ik} \geq r_i - M(1 - y_{ik})$. Optional time windows $[\,\underline{e}_i, \overline{e}_i\,]$ extend this to $\underline{e}_i \leq t^a_{ik} \leq \overline{e}_i$ when $y_{ik} = 1$.

3.6 The dynamic setting

In DDRP the order set $\mathcal{I}$ is not known a priori. Instead orders arrive according to a stochastic process—typically a non-homogeneous Poisson process with rate $\nu(t)$ over a horizon $[0, T]$—and each order becomes *actionable* at its release time r_i. The dispatcher observes the state

$$S_t = (\text{positions and remaining batteries of all drones},\ \text{served orders},\ \text{pending orders})$$

and selects an action: dispatch a drone, insert a customer into an existing route, or idle. A *policy* π maps states to actions; the Dynamic DDRP seeks

$$\min_{\pi}\ \mathbb{E}\Big[\ \sum_{t=0}^{T}\Big(\textstyle\sum_k \sum_{i,j} d_{ij}\, x^t_{ijk}\ +\ \lambda\, U_t\Big)\ \Big], \tag{24}$$

where U_t is the number of still-unserved, already-released orders at epoch t. The expectation is taken over the (unknown) arrival process.

A tractable and widely used approach is *rolling-horizon re-optimisation*: at each decision epoch Δ the dispatcher solves a *static* instance consisting of all pending orders plus new arrivals, with the drones' current positions, remaining payload and remaining battery as initial conditions, and commits to the first portion of the plan before re-optimising at the next epoch. This decomposition turns the online problem into a sequence of offline DDRP instances, each solvable by the trajectory metaheuristics of Section 3.7.

3.7 Trajectory metaheuristics for the DDRP

Encoding. A convenient single-solution encoding is the *giant tour*: a permutation of the orders, split optimally (by dynamic programming) into K feasible drone routes. The split respects capacity (20) and battery (21) and is recomputed after every move. Unserved orders are simply those the split cannot place, and are charged λ in (15).

Neighbourhoods. Classic VRP moves operate on the permutation:

- **relocate** — remove one order and reinsert it elsewhere;
- **exchange** — swap two orders;
- **Or-opt** — move a chain of 2 or 3 consecutive orders;
- **2-opt** — reverse a segment, useful within a route.

Algorithm 3 Tabu search for the (static) DDRP

$s \leftarrow$ CheapestInsertion(pending orders)
$s^\star \leftarrow s$; $\mathcal{T} \leftarrow \emptyset$
while stopping criterion not met **do**
  $s' \leftarrow \arg\min_{\hat{s} \in \mathcal{N}(s)} \{Z(\hat{s}) : \text{move}(\hat{s}) \notin \mathcal{T}\}$
  if s' is tabu but $Z(s') < Z(s^\star)$ **then** ▷ aspiration
    release s' from $\mathcal{T}$
  end if
  $s \leftarrow s'$; insert the reverse move into $\mathcal{T}$ with tenure θ
  if $Z(s) < Z(s^\star)$ **then**
    $s^\star \leftarrow s$
  end if
end while
return $s^\star$

Why trajectory methods suit the dynamic setting. Trajectory methods are the natural choice for DDRP because (i) they are *anytime*, so the dispatcher can halt them when a new order arrives and use the best solution available; (ii) they *warm-start* trivially, by seeding the search with the previous plan so that real-time re-optimisation inherits past effort; and (iii) the giant-tour representation admits cheap, incremental evaluation of relocate/exchange moves. This closes the conceptual circle opened in the introduction: the same skeleton of Algorithm 1 serves both the continuous, multiplication-of-wakes landscape of Part I and the discrete, payload-and-battery-constrained routing of Part II.

4 Synthesis

Table 3 contrasts the two problems. Despite their different mathematical natures—one continuous, one combinatorial—both are constrained, black-box and multimodal, and both are attacked in the MetaLab Suite with the same family of trajectory-based metaheuristics, instantiated with problem-specific encodings and neighbourhoods.

Table 3: Comparison of the two problems.

Aspect	WFLO (Part I)	DDRP (Part II)
Decision	continuous turbine coordinates $X \in \mathbb{R}^{2N}$	discrete routes and assignments
Objective	maximise $\bar{P}(X)$ / AEP	minimise distance $+ \lambda$ (unserved)
Key physics/economics	Jensen wake deficit; wind rose	payload-dependent battery $e(q) = a + bq$
Hard constraints	boundary, exclusion, spacing, slope	capacity Q, battery B, precedence, release times
Source of multimodality	wake coupling; non-convex feasible set	NP-hard routing; online arrivals
Trajectory method	simulated annealing / tabu on coordinates	tabu search / ILS on the giant tour
Special property	derivative-free, nonsmooth objective	*anytime*, warm-startable for re-optimisation

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